

Non-Parametric Spatial Correlation Extraction and a Robust Initialization for the Convex-Concave Procedure

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Abstract—Intra-die process variations in nanometer technologies exhibit spatial correlation that dominates total variation, and statistical timing analysis therefore requires an accurate model of the correlation function. Parametric forms such as the exponential, Gaussian and Matérn families are guaranteed positive definite but impose a presupposed shape and lead to non-convex estimation; non-parametric bases are more flexible but are harder to fit and do not automatically yield a positive-definite covariance. This paper studies non-parametric spatial correlation extraction through a basis expansion of the covariance matrix, fitted by either least squares (LSQ) or maximum likelihood (MLE) with a cutting-plane method. We also derive the multi-chip maximum-likelihood (MLE-M) estimator, including the nugget-modified Matérn function and the concentrated log-likelihood, and compare it with least-squares extraction on synthetic multi-chip data. We first analyze the convexity of the MLE objective: it is a difference of convex functions and is not globally convex. We therefore solve the MLE with the convex-concave procedure (CCP), which linearizes the concave part and solves a sequence of convex subproblems. The main contribution is a robust, dependency-free initialization for the CCP: instead of warm-starting from the least-squares solution whose covariance is only positive semi-definite, we initialize the first majorization matrix from the free (unconstrained) MLE covariance, repaired to the nearest positive-definite matrix by spectral clipping, and return its inverse directly. The resulting initializer is always positive definite, removes the singular-matrix failure mode, and reaches the same optimum as the previous scheme at the cost of two extra iterations. We validate the framework on polynomial and clamped B-spline bases, on isotropic and anisotropic fields, and across Python, C++ and Rust implementations.

I. INTRODUCTION

As semiconductor feature sizes continue to shrink, process variations become a first-order determinant of circuit performance and yield. The traditional corner-based methodology verifies timing at extreme process-voltage-temperature corners, but intra-die variation has grown to dominate the total budget, so corner analysis becomes increasingly pessimistic and expensive. On a representative ten-inverter chain, enforcing timing closure at the worst-case corner required roughly 54% more area and 44% more power than a statistical design that predicts delay at $\mu + 3\sigma$ [1]. Statistical design instead models the process parameters as random variables and propagates their distributions to path delay.

A key ingredient of statistical timing is the spatial correlation of the underlying process parameters. Two devices that are physically close tend to vary together, whereas distant devices are almost independent, and for long paths these correlations dominate the path delay variance. The correlation structure is obtained from measurements on dedicated test chips and is then described by a correlation function of the separation distance. The model must be positive definite, since a covariance matrix built from a non-positive-definite kernel is not a valid covariance matrix.

Two modeling philosophies compete. Parametric kernels—exponential, Gaussian and Matérn—are parameterized by a length scale and are positive definite by construction [2], [3], but they presuppose the shape of the correlation function, and their likelihood is non-convex in the parameters. Non-parametric approaches, such as polynomial and B-spline expansions of the kernel, make no such presupposition and can represent non-monotone correlation functions, but they require a constrained fit and a mechanism to enforce positive definiteness.

This paper develops a non-parametric framework and, within it, identifies and fixes a subtle but critical detail: the initialization of the optimization. Our contributions are as follows.

1. We formulate the extraction as a basis expansion of the covariance matrix and fit it with a cutting-plane method, comparing LSQ and MLE objectives (Section IV).
2. We analyze the convexity of the MLE objective and show it is a difference of convex functions, convex exactly on $0 \prec \Omega \preceq 2Y$ (Section VI).
3. We solve the MLE with the convex-concave procedure (CCP), whose fixed points are stationary points of the true objective (Section VII).
4. We propose a robust initialization for the CCP that replaces the singular-prone least-squares warm start with a guaranteed positive-definite precision obtained from the free MLE and a nearest-positive-definite repair (Section VIII).
5. We validate the approach on polynomial and clamped B-spline bases, on isotropic and anisotropic data, over a range of sample sizes, and across Python, C++ and Rust implementations (Section IX).

II. RELATED WORK

Spatial correlation modeling from silicon measurement data aims to extract the characteristic parameters of a correlation function from a large amount of test data, and it raises two questions: which functional form to use, and how to estimate its parameters [4], [5], [6], [7], [8], [9]. An early approach fitted an exponential correlation with a curve-fitting procedure [7]; a least-squares method recovered the parameters of the Matérn function from data distorted by white noise [6]; and a spectral-domain method exploited the Fourier counterpart of the Matérn function to suppress high-frequency measurement error [9]. A maximum-likelihood method for a *single* test chip was proposed in [8], but it yields one parameter set per chip with no criterion for choosing among them, and it ignores the purely random component and the measurement error. The multi-chip MLE-M method of Section IV addresses all three issues.

The estimated correlation feeds variation-aware design: statistical timing analysis [10], [11], [12], power and leakage minimization [13], [14], and the statistical design methodology that replaces pessimistic corner analysis [1], [15], [16], [17]. The underlying random-field and geostatistical theory is classical [2], [18], [19], [20], [21], [22], [23], [24], as are the maximum-likelihood foundations [25], [26] and the numerical optimizers used to fit them [27], [28].

III. BACKGROUND

A. Process Variation and Spatial Correlation

A process parameter X , such as the effective channel length or the threshold voltage, is modeled as a normal random variable. Its deviation decomposes into an inter-die (die-to-die) component, a spatially correlated intra-die component, and a purely random intra-die component,

$$X = \mu_X + X_{D2D} + X_{WID,c} + X_{WID,r},$$

and, because the components are independent, the variances add:

$$\sigma_X^2 = \sigma_{X_{D2D}}^2 + \sigma_{X_{WID,c}}^2 + \sigma_{X_{WID,r}}^2. \quad (1)$$

Following the conventional decomposition [17], the intra-die part is often refined further into a *deterministic* component, fixed by the layout context and modelable by deliberately exploring layout patterns; a *correlated random* component, random but spatially correlated through proximity effects; and a *purely random* component, which is spatially uncorrelated. We assume the deterministic part to be modeled and removed, and concentrate on the correlated component together with the purely random component and the measurement error [6], [9]. The purely random component and the measurement error produce the nugget effect described below.

The correlation between two gates i and j separated by a distance d_{ij} is often written in the exponential form

$$\rho(X_i, X_j) = K_{D2D} + K_{WID} \exp\left(-\frac{d_{ij}}{CD_{WID}}\right), \quad (2)$$

where K_{D2D} is the inter-die fraction, K_{WID} the correlated intra-die fraction and CD_{WID} the correlation distance [1]. For a path of N gates, the path delay variance is the sum of the gate variances plus the pairwise covariances,

$$\sigma_{DP}^2 = \sum_{i=1}^N \sigma_{D_i}^2 + 2 \sum_{i=1}^N \sum_{j=i+1}^N \text{Cov}(D_i, D_j), \quad (3)$$

and only spatially correlated parameters contribute to the covariance terms. Consequently, correlation dominates the path delay variance as the logic depth grows, which motivates an accurate spatial correlation model.

B. Random Fields and Correlation Functions

A spatially varying quantity is described by a random field $\{\tilde{z}(s) : s \in D\}$, and its second-order structure is captured by the covariance $C(s_i, s_j) = \text{cov}(\tilde{z}(s_i), \tilde{z}(s_j))$ and the correlation

$$R(s_i, s_j) = \frac{C(s_i, s_j)}{\sqrt{C(s_i, s_i) C(s_j, s_j)}}.$$

Under the homogeneous and isotropic assumption (HIF), both depend only on the separation $h = \|s_i - s_j\|$, so that $C(h) = \sigma^2 \rho(h)$. A valid covariance function must be positive definite: for all sites $s_1, \dots, s_n$ and all coefficients $a_1, \dots, a_n$, $\sum_i \sum_j a_i a_j C(s_i, s_j) \geq 0$, with equality only when all $a_i = 0$. Equivalently, by Bochner's theorem, its Fourier transform is a non-negative measure.

C. Parametric and Non-Parametric Models

Common parametric kernels are the exponential $\rho(h) = \exp(-\alpha h)$, the Gaussian $\rho(h) = \exp(-\alpha h^2)$, and the Matérn family

$$\rho(h) = \frac{(\alpha h)^\nu}{2^{\nu-1} \Gamma(\nu)} K_\nu(\alpha h), \quad (4)$$

where K_ν is the modified Bessel function of the second kind and Γ is the gamma function [2]. The Matérn family interpolates between the exponential ($\nu = 1/2$) and the Gaussian (as $\nu \rightarrow \infty$), so it spans a spectrum of smoothness. These kernels are guaranteed positive definite, but estimation is non-convex, the shape is presupposed, and the isotropic assumption may be violated. Non-parametric bases—polynomials and B-splines—remove the shape presupposition and can represent non-monotone correlation, at the cost of a constrained, higher-dimensional fit that does not by itself guarantee positive definiteness.

D. The Nugget Effect and the Modified Matérn Function

Apart from the spatially correlated component, the purely random component of intra-die variation and the unavoidable measurement error also contribute. Both are modeled as Gaussian white noise with a combined variance τ^2 , and they introduce a discontinuity at the origin of the correlation function known as the *nugget effect* [20], [21].

Writing the variance of the correlated component as σ^2 , the modified Matérn correlation function is

$$\tilde{R}(h) = \begin{cases} 1, & h = 0, \\ \frac{\sigma^2}{\sigma^2 + \tau^2} R(h), & h > 0, \end{cases} \quad (5)$$

where $R(h)$ is the Matérn function Eq. 4. The ratio $\kappa = \tau^2/\sigma^2$ measures the noise-to-signal variance, and the modified correlation matrix of the measurements becomes $\tilde{R}(\kappa, \psi) = R(\psi) + \kappa I$ [6], [8]. Ignoring the nugget in the extraction severely distorts the recovered correlation function, as the experiments of Section IX show.

E. Gaussian Process Regression

A Gaussian process is a collection of random variables whose every finite subset is multivariate normal, $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$. For a zero-mean process the kernel k is the spatial correlation function, so fitting a Gaussian process and extracting the correlation are the same problem [3]. The hyperparameters are commonly estimated by minimizing the negative log marginal likelihood with L-BFGS-B and multiple random restarts to mitigate local minima [27]. A Gaussian process trained on a single realization, however, uses only one chip of measurements and is correspondingly noisier than a covariance-based estimator that pools all chips.

IV. PROBLEM FORMULATION

A. Biased Sample Covariance

Let $z_m(s_i)$ denote the measurement at site s_i on chip m , for $m = 1, \dots, M$ and $i = 1, \dots, n$. Removing the across-chip mean at each site gives the centered data $y_m(s_i)$, and the biased sample covariance is

$$Y = \frac{1}{M} \sum_{m=1}^M y_m y_m^\top. \quad (6)$$

When the number of chips is at least the number of sites ($M \geq n$) and the data are sufficiently rich, Y is positive definite; when $M < n$, Y is rank-deficient and singular. This is the statistical reason that “more samples is easier”: the sample covariance approaches the true covariance as M grows, and the estimation problem becomes better conditioned.

B. Basis Expansion

We write the correlation function as a linear combination of basis functions, $\rho(h) = \sum_i p_i \Psi_i(h)$, and form the covariance matrix

$$\Omega(p) = \sum_{k=1}^m p_k F_k, \quad (F_k)_{ij} = \Psi_k(\|s_i - s_j\|), \quad (7)$$

which is affine in the coefficients p . The difficulty is that the MLE objective contains $\Omega(p)^{-1}$, and the inverse of a matrix cannot be expressed in convex form.

C. Least-Squares Estimation

The least-squares (LSQ) estimate minimizes the Frobenius residual to the sample covariance,

$$\min_{p, \kappa} \|\Omega(p) + \kappa I - Y\|_F \quad \text{s.t.} \quad \Omega(p) \succeq 0, \kappa \geq 0, \quad (8)$$

a convex problem whose nugget κI guarantees strict positive definiteness. LSQ targets the shape of the sample covariance and places no upper bound on Ω .

D. Maximum Likelihood Estimation for Multiple Chips

For a single chip, the measurement vector $z = (z(s_1), \dots, z(s_N))$ is N -variate Gaussian with zero mean and covariance $\sigma^2 R$, so its log-likelihood is

$$\log L = -\frac{N}{2} \log 2\pi - \frac{N}{2} \log \sigma^2 - \frac{1}{2} \log \det R - \frac{1}{2\sigma^2} z^\top R^{-1} z. \quad (9)$$

Estimating each chip separately produces one parameter set per chip, with no principled way to choose among them for design. The multi-chip maximum-likelihood (MLE-M) method instead pools all M chips into a single likelihood [8], [29]. Two preprocessing steps align the data with the measurement process. First, the inter-die component is removed by subtracting the per-site across-chip mean,

$$z_m^*(s_i) = z_m(s_i) - \frac{1}{M} \sum_{k=1}^M z_k(s_i). \quad (10)$$

Second, the nugget is incorporated through the modified correlation matrix $\tilde{R}(\kappa, \psi) = R(\psi) + \kappa I$ with $\kappa = \tau^2/\sigma^2$. The likelihood of all M chips is then

$$\log L(\sigma^2, \kappa, \psi) = -\frac{MN}{2} \log 2\pi - \frac{MN}{2} \log \sigma^2 - \frac{M}{2} \log \det \tilde{R} - \frac{1}{2\sigma^2} \sum_{m=1}^M z_m^* z_m^{*\top} \quad (11)$$

Setting $\partial \log L / \partial \sigma^2 = 0$ gives the variance estimate

$$\hat{\sigma}^2(\tilde{R}) = \frac{1}{MN} \sum_{m=1}^M z_m^{*\top} \tilde{R}^{-1} z_m^* = \frac{1}{N} \text{tr}(Y \tilde{R}^{-1}), \quad Y = \frac{1}{M} \sum_{m=1}^M z_m^* z_m^{*\top} \quad (12)$$

where the last step uses $\text{tr}(AB) = \text{tr}(BA)$. Substituting Eq. 12 into Eq. 11 and dropping constants yields the concentrated log-likelihood

$$\log L_0(\kappa, \psi) = -\log \det \tilde{R} - N \log(\text{tr}(Y \tilde{R}^{-1})), \quad (13)$$

which is optimized numerically over (κ, ψ) and followed by back substitution for $\hat{\sigma}^2$ [25]. In practice, $\log \det \tilde{R}$ is evaluated from an LU factorization, $\log \det \tilde{R} = \sum_{j=1}^N \log |u_{jj}|$, because forming the determinant directly can underflow to a “log of zero” error; the outer problem is solved by a standard nonlinear optimizer [27]. Algorithm 1 summarizes the estimator.

MLE-M estimates the parameters of a parametric kernel. The non-parametric framework of this paper uses the same sample covariance Y of Eq. 12 but replaces $R(\psi)$ by the basis expansion Eq. 7, so that the concentrated likelihood Eq. 13 becomes a function of the coefficients p . As shown in Section VI, this objective is a difference of convex functions, which is solved by the CCP of Section VII.

Algorithm 1 Multi-chip maximum likelihood (MLE-M)

Require: measurements $\{z_m\}_{m=1}^M$, sites $\{s_i\}$, ratio κ

- 1: $z_m^*(s_i) \leftarrow z_m(s_i) - \frac{1}{M} \sum_{k=1}^M z_k(s_i)$
- 2: $Y \leftarrow \frac{1}{M} \sum_{m=1}^M z_m^* z_m^{*\top}$
- 3: $\tilde{R}(\kappa, \psi) \leftarrow \tilde{R}(\psi) + \kappa I$
- 4: $g \leftarrow \log \det \tilde{R} + N \log \text{tr}(Y \tilde{R}^{-1})$
- 5: $(\kappa^*, \psi^*) \leftarrow \min_{\kappa, \psi} g$
- 6: $\hat{\sigma}^2 \leftarrow \frac{1}{N} \text{tr}(Y \tilde{R}^{-1})$
- 7: **return** $\hat{\sigma}^2, \kappa^*, \psi^*$

V. NON-PARAMETRIC BASES

A. Polynomial Basis and Conditioning

The polynomial basis uses the monomials of the distance, $\varphi_k(d) = d^k$ for $k = 0, \dots, m-1$. It is simple and universal, but every basis function has global support, so the columns of the design matrix

$$A = [\text{vec}(F_1) \cdots \text{vec}(F_m)] \quad (14)$$

become highly correlated as m grows. The condition number $\text{cond}(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$ then grows rapidly: on a 20-site problem it increases from 1.2×10^1 at $m = 2$ to 3.5×10^{10} at $m = 10$, that is, by nine orders of magnitude.

B. Clamped B-Spline Basis

The B-spline basis uses quadratic ($k = 2$) basis functions B_i on a knot vector t ,

$$K(d) = \sum_{i=1}^m c_i B_i(d), \quad B_i \geq 0, \quad \sum_i B_i = 1, \quad (15)$$

each with local support of roughly three knot spans. The knot vector must span the *actual* data domain $[0, d_{\max}]$; a common error is to spread uniform knots over a much smaller interval, so that most basis values are extrapolated polynomials outside the valid domain and are numerically meaningless. Clamped knots over the data domain repair this, as Fig. 1 illustrates.

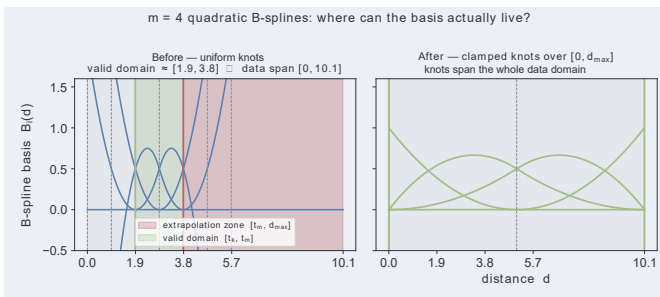


Figure 1: The legacy uniform knots span only a fraction of the data domain, so most basis values are meaningless extrapolations.

The clamped knot vector repeats the end knots $k+1$ times,

$$t = \underbrace{[0, \dots, 0]}_{k+1} \cup \{\text{interior of } \text{linspace}(0, d_{\max}, m-k+1)\} \cup \underbrace{[d_{\max}, \dots, d_{\max}]}_{k+1} \quad (16)$$

Algorithm 2 Clamped quadratic B-spline basis

Require: sites $\{s_i\}$, control points m , degree $k = 2$

- 1: **if** $m < k + 1$ **then**
- 2: **raise** VALUEERROR
- 3: **end if**
- 4: $d_{\max} \leftarrow \max_{i,j} \|s_i - s_j\|$
- 5: $g \leftarrow \text{linspace}(0, d_{\max}, m - k + 1)$
- 6: $t \leftarrow [0]_{k+1} \parallel \text{interior}(g) \parallel [d_{\max}]_{k+1}$
- 7: $F_i \leftarrow B_{i,k}(t)$ on the pairwise distance matrix
- 8: **return** $\{F_i\}, t, k$

which confines every basis element to the data span. On the same 20-site problem the condition number of the clamped B-spline design matrix stays near 5 to 7 across $m = 4, \dots, 10$, four orders of magnitude better than the polynomial basis at $m = 6$. Algorithm 2 lists the construction.

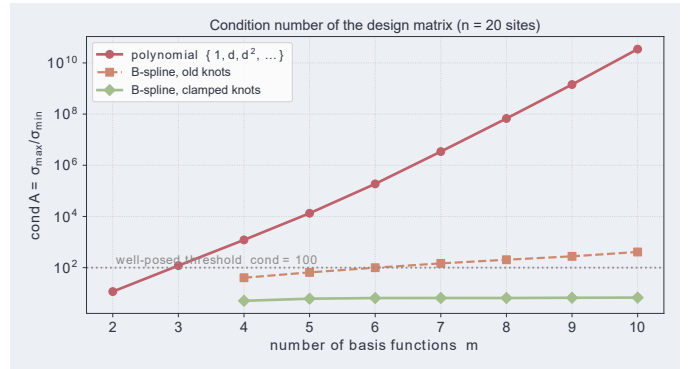


Figure 2: Design-matrix condition numbers for the polynomial, clamped B-spline and legacy B-spline bases.

Monotonicity of the fitted kernel is imposed by constraining the control coefficients to be non-increasing. Because the quadratic B-spline basis functions are non-negative and sum to one, a non-increasing coefficient sequence yields a non-increasing spline, which is the desired behavior for a correlation function. The monotonicity constraint must be applied to *exactly* the control coefficients; mixing it with an objective variable changes the meaning of the index and silently frees or fixes the wrong coefficients.

VI. CONVEXITY OF THE MAXIMUM-LIKELIHOOD OBJECTIVE

A. A Difference of Convex Functions

The MLE objective without the nugget is

$$f(\Omega) = \underbrace{\text{Tr}(\Omega^{-1}Y)}_{\text{convex } h} + \underbrace{\log \det \Omega}_{\text{concave}}, \quad (17)$$

a difference of convex functions: $\text{Tr}(\Omega^{-1}Y)$ is convex in Ω , while $\log \det \Omega$ is concave, equivalently $-\log \det \Omega$ is convex. A difference of two convex functions is, in general, neither convex nor concave, so the ellipsoid method cannot be applied to Eq. 17 directly.

B. The Convexity Region

The second directional derivative of f along a symmetric direction U , with $A = \Omega^{-1/2}U\Omega^{-1/2}$ and $B = \Omega^{-1/2}Y\Omega^{-1/2}$, is

$$D^2f(U) = \text{Tr}(A^2(2B - I)) \succeq 0 \iff 2B \succeq I \iff \Omega \preceq 2Y. \quad (18)$$

Hence f is convex exactly on the region $\Delta_{2Y} = \{\Omega : 0 \prec \Omega \preceq 2Y\}$, with genuine negative curvature outside it. This region coincides with the classical result that the Gaussian log-likelihood is strictly concave “in and only in” Δ_{2S_n} [30]. Along the commuting ray $\Omega = tY$ the objective collapses to one variable,

$$f(t) = n \log t + \log \det Y + \frac{n}{t}, \quad f''(t) = \frac{n(2-t)}{t^3}, \quad (19)$$

which is convex for $t < 2$ and concave for $t > 2$, with the sign flip exactly at $\Omega = 2Y$. Fig. 3 visualises the sign flip.

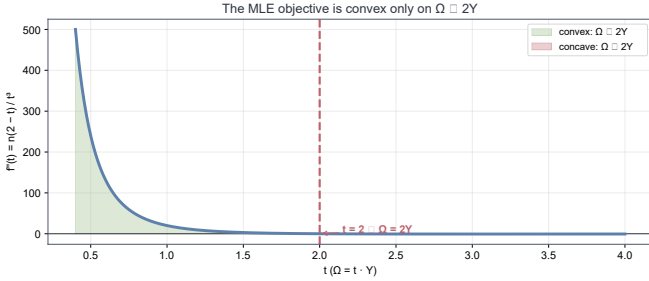


Figure 3: Second derivative of the objective along the commuting ray $\Omega = tY$; it changes sign exactly at $t = 2$.

VII. THE CONVEX-CONCAVE PROCEDURE

A. Majorization

The convex-concave procedure (CCP) handles a difference of convex functions by linearizing the concave part and solving a sequence of convex problems. Since $\log \det \Omega$ is concave, its tangent at the current iterate Ω_k is a global upper bound:

$$\log \det \Omega \leq \log \det \Omega_k + \text{Tr}(\Omega_k^{-1}(\Omega - \Omega_k)). \quad (20)$$

Dropping constants and retaining the convex terms gives the surrogate

$$S_k(\Omega) = \text{Tr}(\Omega^{-1}Y) + \text{Tr}(\Omega_k^{-1}\Omega), \quad (21)$$

which is convex in Ω .

B. Majorize-Minimize and Stationary Points

Each round minimizes the surrogate over the basis family; the chain

$$f(\Omega_{k+1}) \leq S_k(\Omega_{k+1}) \leq S_k(\Omega_k) = f(\Omega_k) \quad (22)$$

guarantees monotone decrease of the objective. The gradient of the surrogate at Ω_k equals the gradient of f there, so a fixed point of the CCP is a stationary point of the true maximum-likelihood problem. The CCP therefore recovers the unconstrained family maximum-likelihood estimate. Fig. 4 shows the CCP acting on a monotone B-spline fit.

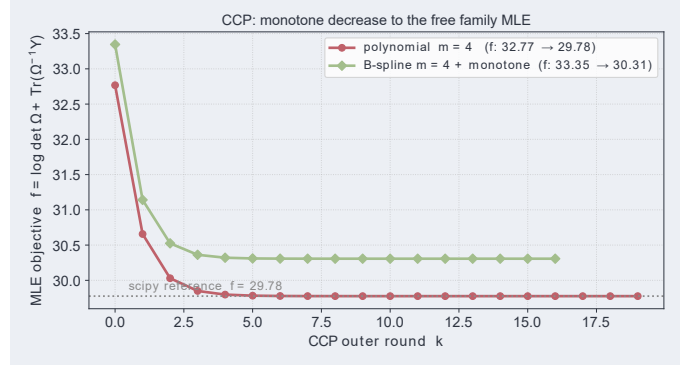


Figure 4: CCP with a monotone B-spline basis: the objective decreases monotonically toward the free family maximum-likelihood estimate.

Algorithm 3 CCP for the family-constrained MLE

Require: Y , basis $\{F_k\}$, start x_0 , round limit N , tolerance τ

- 1: $x \leftarrow x_0$; $M \leftarrow \text{MLECORRMtx}(Y)$
- 2: **for** $k \leftarrow 0$ **to** $N - 1$ **do**
- 3: $O \leftarrow \text{CCPORACLE}(\{F_k\}, Y, M)$
- 4: $x' \leftarrow \text{CUTTINGPLANE}(O, x)$
- 5: **if** $x' = \text{none}$ **then**
- 6: **return** x
- 7: **end if**
- 8: **if** $|f(x') - f(x)| < \tau$ **then**
- 9: **return** x'
- 10: **end if**
- 11: $x \leftarrow x'$; $M \leftarrow \Omega(x)^{-1}$
- 12: **end for**
- 13: **return** x

C. The Majorization Matrix

The surrogate Eq. 21 depends on the iterate only through the matrix $M_k = \Omega_k^{-1}$:

$$S_k(\Omega) = \text{Tr}(\Omega^{-1}Y) + \text{Tr}(M_k\Omega), \quad M_k = \Omega_k^{-1}. \quad (23)$$

Every round of the CCP therefore requires the inverse of the current iterate, and the first round requires an *initial* majorization matrix $M_0 = \Omega_0^{-1}$. Algorithm 3 summarizes the procedure.

VIII. INITIALIZING THE CONVEX-CONCAVE PROCEDURE

A. The Problem: Positive Semi-Definite Is Not Positive Definite

The original implementation warm-started the CCP from the least-squares solution x_0 , computing the first majorization matrix as $M_0 = \Omega(x_0)^{-1}$. The LSQ oracle enforces only $\Omega(x_0) \succeq 0$, the *closed* positive semi-definite cone, whose boundary is allowed. The CCP majorization, however, requires $\Omega_0 \succ 0$ so that Ω_0^{-1} exists and $\log \det \Omega_0$ is finite. If $\Omega(x_0)$ is singular the matrix inverse raises an error and the solver fails at the first iteration. A concrete failure occurs on the polynomial basis with $F_0 = \mathbf{1}$ (the all-ones matrix): the start $x_0 = [4, 0, 0, 0]$ gives $\Omega(x_0) = 4 \cdot \mathbf{1}$, which is rank one and not invertible. Even a near-singular $\Omega(x_0)$ yields a numerically worthless majorization matrix. In short, one cannot guarantee that the least-squares start yields a positive-definite covariance.

Algorithm 4 Initial majorization matrix MLECORRMTX

Require: sample covariance Y , relative floor ε

- 1: $Y \leftarrow (Y + Y^\top)/2$
- 2: $(w, V) \leftarrow \text{EIGH}(Y)$ ▷ ascending eigenvalues
- 3: $\delta \leftarrow \varepsilon \cdot \max(w_{\max}, 1)$
- 4: $w \leftarrow \max(w, \delta)$ ▷ nearest-PD spectrum
- 5: **return** $(V \cdot (1/w)) V^\top$ ▷ Ω_0^{-1} , directly

B. The Free Maximum-Likelihood Covariance

To obtain a guaranteed positive-definite start, consider the unconstrained maximum-likelihood problem, obtained by dropping the basis constraint and minimizing over all positive-definite matrices,

$$\min_{\Omega \succ 0} \log \det \Omega + \text{Tr}(\Omega^{-1}Y). \quad (24)$$

Setting the gradient to zero, $\Omega^{-1} - Y = 0$, gives the closed-form solution

$$\Omega^* = Y. \quad (25)$$

The unconstrained maximum-likelihood covariance is simply the sample covariance Y itself. This is the natural initialization: it is the exact minimizer of the objective without the family constraint, so the first majorization is performed at the unconstrained optimum.

C. Nearest Positive-Definite Repair

The free problem Eq. 24 is unbounded when Y is not positive definite: taking $S = I + tvv^\top$ with $v \in \text{null}(Y)$ leaves $\text{Tr}(SY)$ constant while $\log \det S \rightarrow \infty$. When Y is not positive definite we therefore repair it to the nearest positive-definite matrix in Frobenius norm, which is obtained by clipping its eigenvalues [31]. Writing the symmetric eigendecomposition $Y = V \text{diag}(\lambda) V^\top$, the repaired covariance is

$$\Omega_0 = V \text{diag}(\max(\lambda, \varepsilon)) V^\top, \quad (26)$$

where ε is a small positive floor relative to $\lambda_{\max}$. The result is positive definite by construction, hence always invertible.

D. Returning the Precision Directly

The CCP never needs Ω_0 itself—only its inverse. Reciprocating the clipped eigenvalues avoids forming and inverting a potentially ill-conditioned matrix:

$$\Omega_0^{-1} = V \text{diag}\left(\frac{1}{\max(\lambda, \varepsilon)}\right) V^\top. \quad (27)$$

This yields the initializer `mle_corr_mtx`, which computes the free-MLE precision in a single symmetric eigendecomposition and requires no external semidefinite-programming solver:

For a positive-definite Y the floor is inactive and the initializer is exactly the free-MLE precision Y^{-1} . The CCP algorithm then becomes: initialize $M_0 = \text{mle_corr_mtx}(Y)$; for $k = 0, 1, \dots$ set $M_k = M_0$ if $k = 0$ and $M_k = \Omega(x_k)^{-1}$ otherwise; solve the convex subproblem Eq. 21; and stop when the objective stalls.

IX. NUMERICAL EXPERIMENTS

A. Setup

Throughout, the site layout is a Halton low-discrepancy grid, and the data are generated from a known kernel so that the fitted model can be compared with the ground truth. Unless stated otherwise, $n = 20$ sites and $N = 3000$ samples are used, and the isotropic generating kernel is $K(d) = 4 \exp(-0.12d^2)$. The cutting-plane solver is shared by all bases; the B-spline fits additionally use the monotone coefficient oracle. The anisotropic experiments use per-axis length scales ℓ_1, ℓ_2 and the four-parameter kernel $k = \sigma^2 \exp(-\frac{1}{2}[(dx)^2/\ell_1^2 + (dy)^2/\ell_2^2])$.

B. Basis Conditioning and Knots

Table I compares the design-matrix condition numbers of the polynomial basis, the clamped B-spline basis, and a B-spline basis built on legacy uniform knots that span only $[0, 1.2\|s_n - s_1\|]$ rather than the data domain. The polynomial basis degrades by nine orders of magnitude; the legacy B-spline stays in the hundreds and extrapolates over most of the data domain; the clamped B-spline remains near 5 to 7.

With the legacy knots the fitted kernel is non-monotone at 268 to 324 grid points; with clamped knots and the monotone coefficient constraint it is monotone at every point. At comparable accuracy, the clamped B-spline basis is four orders of magnitude better conditioned, so sane conditioning is available at no cost in fidelity.

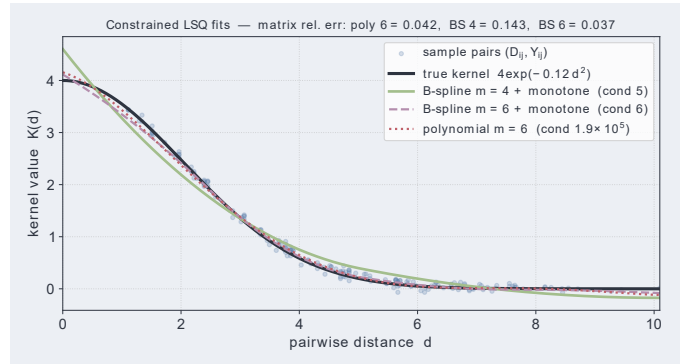


Figure 5: Fitted kernels versus the true kernel for the polynomial and clamped B-spline bases.

C. Least Squares versus Maximum Likelihood versus CCP

Table II reports the relative error against the true kernel for LSQ, the plain MLE oracle, and the CCP. The LSQ fit tracks the true kernel closely; the MLE oracle is pinned to its bound and its covariance is nearly singular, so it does not track the kernel; the CCP recovers the unconstrained family MLE.

The gap between the MLE and the true kernel does not close as the sample size grows: even as $N \rightarrow \infty$, $Y \rightarrow \Sigma_{\text{true}}$, yet the family projection remains outside $\Delta_{2\Sigma_{\text{true}}}$ by about 1.0. The failure is structural—model misspecification—not

Table I: Design-matrix condition numbers $\text{cond}(A)$ for the three bases.

m	polynomial	B-spline (clamped)	B-spline (legacy knots)
4	1.21×10^3	5.05	40.26
6	1.87×10^5	6.48	99.40
8	6.72×10^7	6.51	203.57
10	3.45×10^{10}	6.75	409.47

Table II: Relative error against the generating kernel. The MLE oracle is pinned to its bound; the CCP recovers the free family MLE.

problem	LSQ rel-err	MLE rel-err	CCP rel-err	CCP objective
iso (1, 1)	0.128	0.705	0.293	29.78
aniso (1, 3)	0.334	0.451	0.451	—
aniso (3, 1)	0.476	0.716	0.683	—

statistical. This is the practical meaning of “the more you know, the easier”: an incorrect parametric model can be as harmful as too few samples.

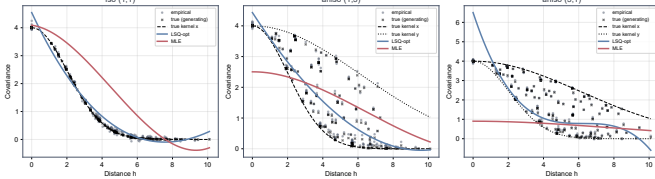


Figure 6: Fitted correlation functions versus the generating covariance for the isotropic and anisotropic problems.

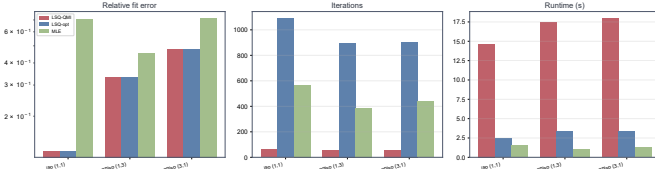


Figure 7: Relative fit error, iteration count and runtime for the LSQ and MLE solvers.

D. Multi-Chip Likelihood: MLE-M, MLEsim and RESCF

The MLE-M method of Section IV was compared with two alternatives on synthetic data: MLEsim, which ignores the nugget and uses the unmodified $R(h)$ in the likelihood, and RESCF, a least-squares-based extraction [6], [9]. The synthetic data contained four components—the spatially correlated component, the purely random component, inter-die variation, and measurement error—generated by the Cholesky method [19]. Two sampling schemes were used: uniform gridding and Monte Carlo. Table III reports the relative error of the variance $\text{err}(\sigma^2)$ and of the correlation function $\text{err}(R(h))$ averaged over thirty runs, for $M \in \{500, 1000\}$ chips, $N \in \{11 \times 11, 21 \times 21\}$ sites, and nugget levels $\kappa \in \{10\%, 50\%, 100\%\}$. Ignoring the nugget (MLEsim) leaves errors above 30% and up to 78%; incorporating it (MLEnug) reduces the correlation-function error below 2.3% in every case and outperforms the least-squares baseline with lower runtime.

E. Robust Initialization of the CCP

Table IV compares the previous least-squares warm start with the free-MLE initializer of Section VIII on the standard 20-site, $m = 4$ problem. Both reach the identical objective 29.775870, with a coefficient difference of 1.9×10^{-4} . The new initializer uses two additional linearization rounds, but it tolerates a singular start that previously raised an error, converging to the same optimum. The initialization therefore changes the optimization path, not the answer.

The initializer is validated by unit tests: for positive-definite Y it equals $\text{inv}(Y)$; for indefinite Y it returns a positive-definite precision whose inverse is positive definite; and the CCP accepts a singular start. The full test suite passes without new dependencies, using only a symmetric eigendecomposition.

F. Sample-Size Sweeps

Fig. 8 shows the relative error of the fitted kernel against the number of samples N for the polynomial and B-spline bases; Fig. 9 and Fig. 10 show the behavior of the LSQ, MLE and CCP solvers over $N = 5, \dots, 2000$ and the fine range $N = 1, \dots, 200$. For $M \geq n$ the length-scale error falls rapidly and flattens beyond about $M \approx 200$ chips, consistent with a covariance-estimation error proportional to $1/\sqrt{M}$. For $M < n$ the sample covariance is rank-deficient and the MLE oracle is infeasible, whereas the monotone B-spline CCP stays well-posed because the monotone constraint keeps the covariance away from the singular boundary. Across $N = 1, \dots, 50$ the clamped B-spline fit is monotone at every point, while the unconstrained polynomial fit is non-monotone at most intermediate sample sizes.

Table V shows the CCP at small sample sizes. The procedure decreases the objective monotonically at every N , even at the rank-deficient $N = 1$, and the monotonicity penalty (the difference between the B-spline and polynomial final objectives) shrinks from 4.7 at $N = 1$ to 0.26 at $N = 50$.

G. Anisotropy and Kernel Misspecification

Two further experiments probe robustness. Fig. 11 compares the true and fitted correlation functions obtained by Gaussian process regression on a single chip [3] and

Table III: Relative-error ranges for multi-chip spatial correlation extraction. The nugget-aware MLE-M (MLEnug) dominates both alternatives.

scheme	method	$\text{err}(\sigma^2)$	$\text{err}(R(h))$
uniform gridding	RESCF (LSE)	0.95–7.31%	0.75–4.26%
uniform gridding	MLEsim (no nugget)	10.98–99.18%	5.44–55.72%
uniform gridding	MLEnug	0.30–4.12%	0.27–2.27%
Monte Carlo	RESCF (LSE)	0.91–2.32%	1.06–2.33%
Monte Carlo	MLEsim (no nugget)	7.58–86.13%	42.48–77.92%
Monte Carlo	MLEnug	0.32–1.67%	0.29–1.39%

Table IV: Effect of the CCP initialization. The new scheme is robust to singular starts at the same optimum.

	LSQ warm start (old)	free-MLE init (new)
CCP objective f	29.775870	29.775870
coefficient rel. difference	—	1.9×10^{-4}
linearization rounds	16	18
singular $\Omega(x_0)$	error	works

Table V: CCP at small sample sizes: initial and final objective with the number of linearization rounds in parentheses.

N	polynomial $f_0 \rightarrow f_1$ (rounds)	monotone B-spline $f_0 \rightarrow f_1$ (rounds)
1	64.14 $\rightarrow$ 25.92 (28)	55.30 $\rightarrow$ 30.62 (41)
5	39.36 $\rightarrow$ 26.92 (30)	33.34 $\rightarrow$ 27.35 (20)
10	41.54 $\rightarrow$ 29.69 (21)	36.35 $\rightarrow$ 29.73 (16)
20	35.69 $\rightarrow$ 26.92 (21)	30.30 $\rightarrow$ 27.01 (16)
50	35.82 $\rightarrow$ 28.43 (19)	33.00 $\rightarrow$ 28.69 (16)

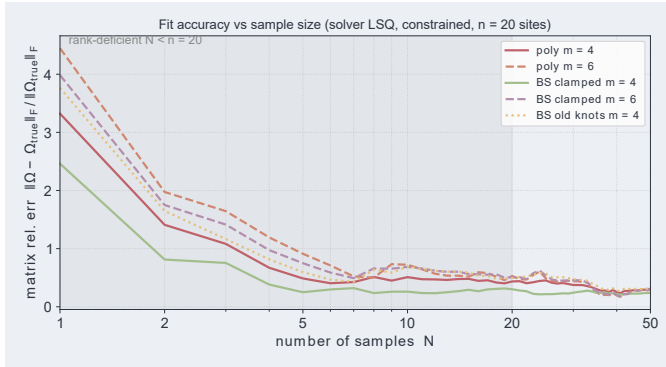


Figure 8: Relative error versus the number of samples N for the polynomial and clamped B-spline bases.

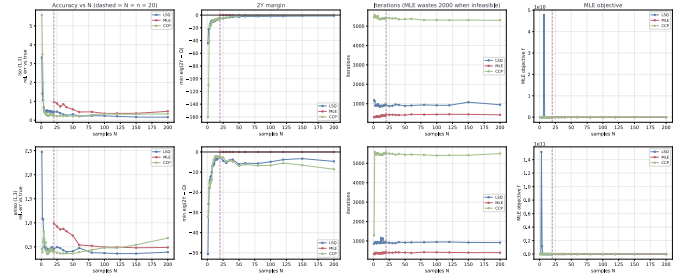


Figure 10: Fine sample-size sweep over $N = 1, \dots, 200$.

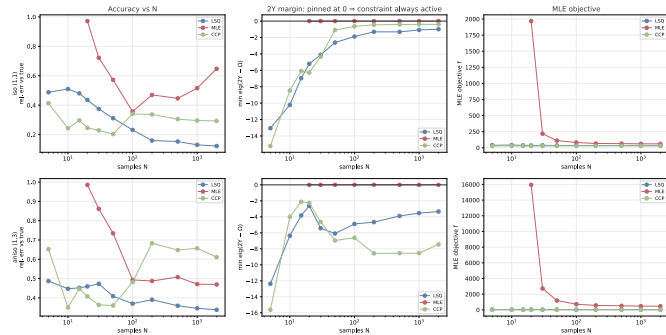


Figure 9: Accuracy and objective of the LSQ, MLE and CCP solvers across sample sizes.

by the multi-realization MLE at $N = 200$; both track the true curve closely when the kernel is correct. Fig. 12 shows that the MLE length-scale error falls with the number of chips, confirming that 100 to 200 chips suffice for reliable extraction. Kernel misspecification, however, is not cured by more data: Fig. 13 is the matrix of length-scale errors obtained when data from one kernel is fitted with another. The diagonal is the correct kernel; fitting a smooth (Gaussian) kernel with a rough (exponential) one inflates the estimated length scale many-fold, whereas exponential data is fitted robustly by the other kernels, and the Matérn-3/2 kernel is the safest default when the process is unknown. Table VI reports the anisotropic extraction accuracy at $N = 200$: both length scales are recovered to within about one percent for the Gaussian and Matérn kernels and a few percent for the exponential. Fig. 14 shows the anisotropic correlation surfaces $C(dx, dy)$; the elliptical contours reveal the directional dependence, and assuming isotropy for such data yields a compromise length scale that misrepresents both directions.

Table VI: Anisotropic length-scale extraction accuracy at $N = 200$.

kernel	true (ℓ_1, ℓ_2)	est. ℓ_1	est. ℓ_2	σ error
Gaussian	(1.0, 4.0)	1.010	4.010	0.020
Gaussian	(3.0, 1.0)	2.992	1.004	0.002
Gaussian	(1.5, 6.0)	1.503	6.010	0.013
Matérn-3/2	(1.0, 4.0)	1.009	4.018	0.011
Matérn-5/2	(4.0, 1.0)	3.978	1.008	0.003
Exponential	(2.0, 6.0)	1.950	5.845	0.015

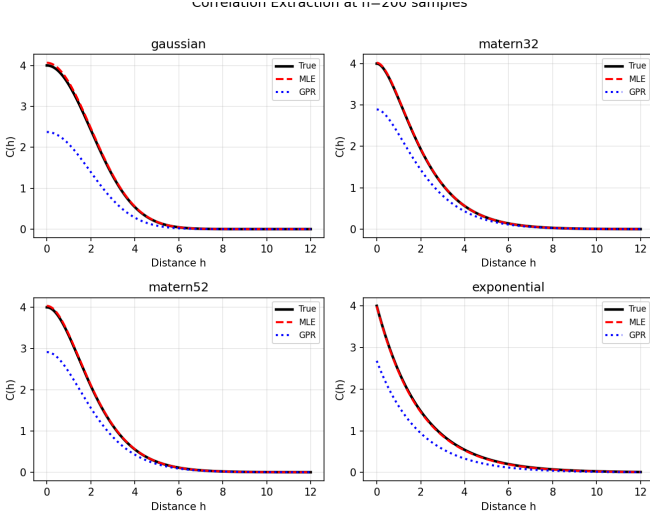
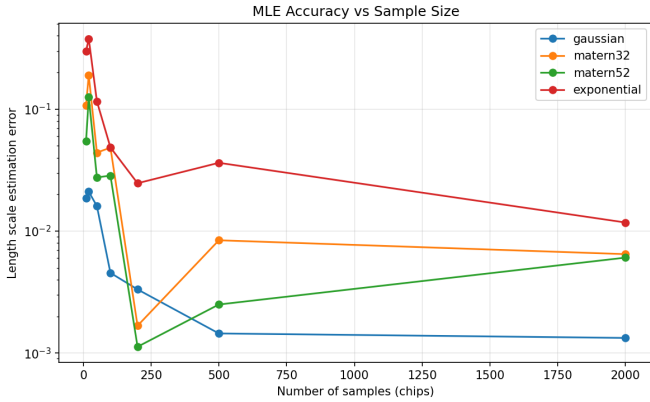
Figure 11: True versus fitted correlation functions for Gaussian process regression and multi-realization maximum likelihood at $N = 200$.

Figure 12: MLE length-scale error versus the number of samples (chips).

H. Cross-Language Verification

The algorithms were reimplemented in Python, C++ and Rust. The conditioning values agree to nine significant digits, and the reference condition number at $m = 10$ was adjudicated at sixty digits. Two cross-language bugs were found and fixed: a triangular inverse computed with a routine that assumes a symmetric matrix, which silently returned $\text{diag}(1/R_{ii})$ instead of R^{-1} and corrupted both the objective and the gradient of the MLE; and the monotonicity index arithmetic, which assumed a trailing

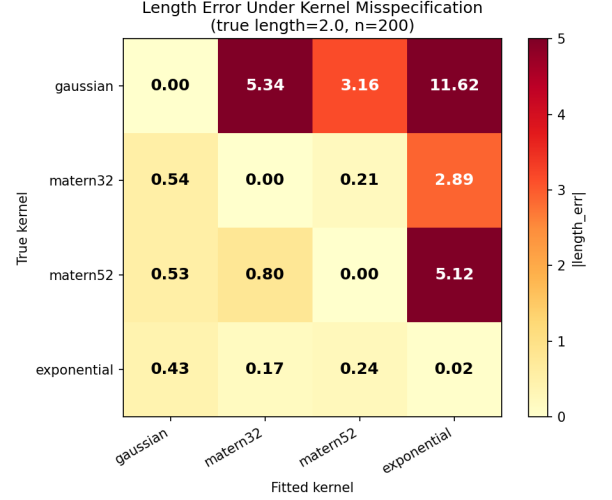
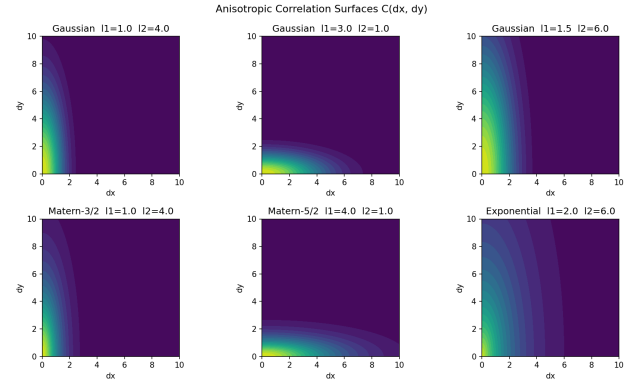


Figure 13: Length-scale error when data generated from one kernel is fitted with another; the diagonal is the correct kernel.

Figure 14: Anisotropic correlation surfaces $C(dx, dy)$; the elliptical contours reveal the directional dependence.

objective variable that only one solver core carries. The three implementations now agree on every reported quantity.

X. DISCUSSION

The initializer hardens only the first round of the CCP. Subsequent rounds invert the current family iterate $\Omega(x_k)$, which can in principle approach the positive semi-definite boundary; the lecture formulation avoids this by including a nugget, $\Sigma = \Omega(p) + \kappa I$ with $\kappa \geq 0$, which keeps Σ positive definite at every round. Adding such a nugget to the solver is a natural extension.

The framework has several further limitations. Non-parametric bases with global support suffer severe conditioning unless the basis is local (B-spline) or small; a monotone fit is not necessarily non-negative, so the minimum eigenvalue of the fitted covariance should be checked separately; and the LSQ and MLE objectives answer different questions, so the relative error against the generating kernel is not a fair criterion for the MLE, which optimizes the likelihood rather than the shape. Finally, when the parametric form is known, fitting that form remains the simplest and most accurate option; the non-parametric machinery is for the case where the shape of the correlation function is unknown.

XI. CONCLUSION

We studied non-parametric spatial correlation extraction as a basis expansion of the covariance matrix, fitted by a cutting-plane method under LSQ or MLE objectives. The MLE objective is a difference of convex functions and is not globally convex, so we solve the problem with the convex-concave procedure, whose fixed points are stationary points of the true objective. Our main contribution is a robust initialization for the procedure. Rather than warm-starting from the least-squares solution, whose covariance is only positive semi-definite, we initialize the first majorization matrix from the free maximum-likelihood covariance, repaired to the nearest positive-definite matrix by spectral clipping, and return its inverse directly. The initializer is always positive definite, eliminates the singular-matrix failure mode, and reaches the same optimum with two extra iterations. The approach is validated on polynomial and clamped B-spline bases, on isotropic and anisotropic fields, over a range of sample sizes, and across Python, C++ and Rust implementations.

XII. CODE AVAILABILITY

The reference implementation, together with the experiments and the cross-language ports, is available in the `corr-solver` project (<https://github.com/luk036/corr-solver>).

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